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Problem 1 :

Problem 1

$$Y|\theta \sim \text{NegBinomial}(\theta, m)$$
$$\text{PMF: } \binom{x+m-1}{x} (1-\theta)^m \theta^x \rightarrow \propto (1-\theta)^m \theta^x$$
$$\theta \sim \text{Beta}(a, b)$$
$$\text{PDF: } \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} \rightarrow \propto \theta^{a-1} (1-\theta)^{b-1}$$

Posterior

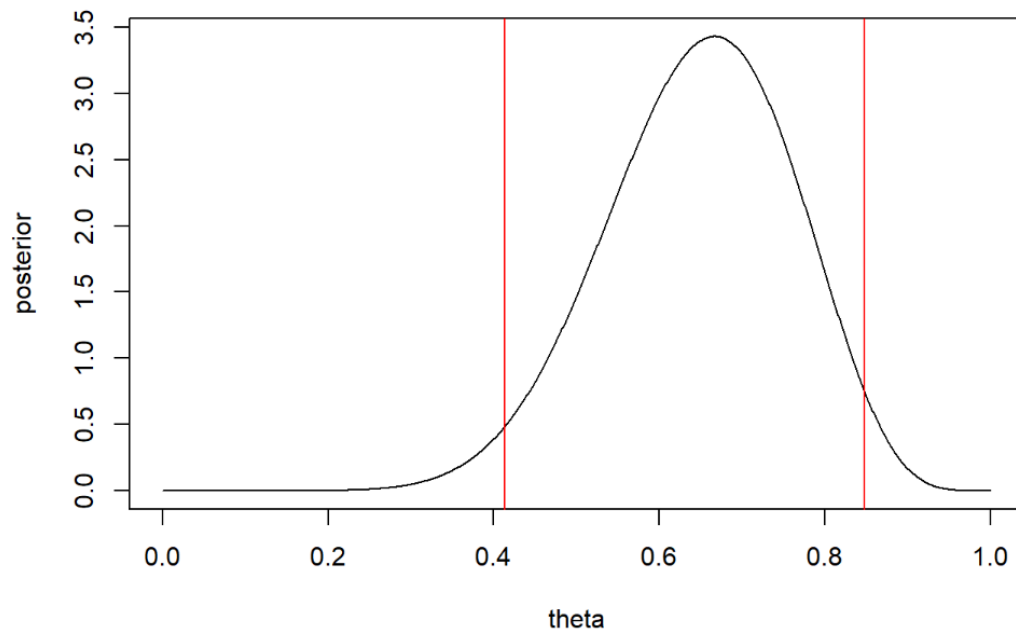
$$P(\theta|Y) \propto P(Y|\theta) P(\theta)$$
$$[(1-\theta)^m \theta^Y] \cdot [\theta^{a-1} (1-\theta)^{b-1}]$$
$$\theta^{Y+a-1} (1-\theta)^{m+b-1} \rightarrow \text{Let: } A = Y+a$$
$$\propto \theta^{A-1} (1-\theta)^{B-1} \rightarrow \text{Beta PDF} \quad B = m+b$$
$$\theta|Y \sim \text{Beta}(Y+a, m+b)$$

Problem 1

```
## Plot the distribution
theta <- seq(0, 1, by = 0.001)
y <- 10
m <- 5
a <- b <- 1
posterior <- dbeta(theta, y+a, m+b)

## Credible intervals
ci <- qbeta(c(0.025, 0.975), y+a, m+b) #95% credible interval
ci_lo <- ci[1]
ci_up <- ci[2]

plot(theta, posterior, type = 'l')
abline(v = c(ci_lo, ci_up), col = c("red", "red"))
```



Problem 2 :

$\theta | Y \sim \text{Beta}(\gamma + a, m + b)$
 Problem 2
 $Y | \lambda \sim \text{Poisson}(\lambda)$
 PMF: $\frac{(e^{-\lambda}) (\lambda)^Y}{Y!} \rightarrow \propto e^{-\lambda} \lambda^Y$ $\uparrow \exp(-\lambda)$
 $\lambda \sim \text{Uniform}(0, 100)$
 PMF: $\frac{1}{b-a}$
 Posterior
 $P(\theta | Y) \propto P(Y | \lambda) P(\lambda)$
 $\propto e^{-\lambda} \lambda^Y \rightarrow \text{Poisson PMF}$
 $\lambda | Y \sim \text{Poisson}(\lambda)$

Problem 2

```
# Define the parameters
lambda <- 20 # mean number of events
x <- 0:100   # X-axis values

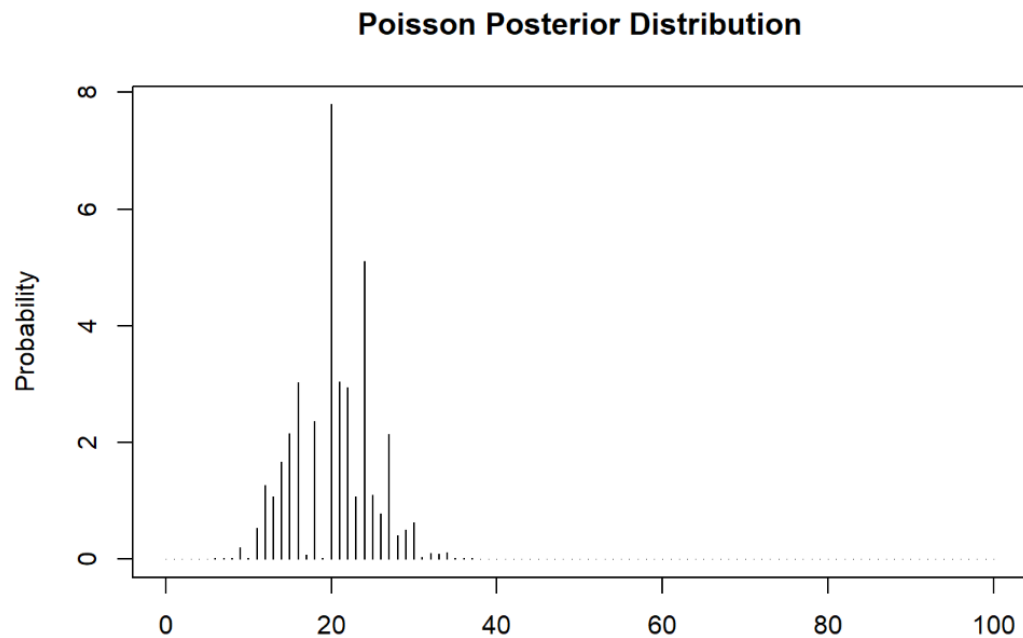
# Calculate the Poisson probability mass function
likelihood <- dpois(x, lambda)

# Define the prior as a uniform distribution
prior <- runif(100, min = 0, max = 100)

# Calculate the unnormalized posterior
unnorm_posterior <- likelihood * prior
```

```
## Warning in likelihood * prior: longer object length is not a multiple of
## shorter object length
```

```
# Plot the posterior distribution
plot(x, unnorm_posterior, type = 'h', main = "Poisson Posterior Distribution", xlab = "X", ylab = "Probability")
```



Problem 3 :

Problem 3

$$Y|\theta \sim \text{Binomial}(N, \theta)$$

$$\text{PMF} = \binom{N}{Y} \theta^Y (1-\theta)^{N-Y}$$

$$\theta \sim \text{Uniform}(0,1) = \theta \sim \text{Beta}(1,1)$$

$$\text{PDF} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$$

$$P(\theta|Y) \propto P(Y|\theta) P(\theta)$$

$$\propto [\theta^Y (1-\theta)^{N-Y}] [\theta^{a-1} (1-\theta)^{b-1}]$$

$$\theta^{Y+a-1} (1-\theta)^{N-Y+b-1} \rightarrow \text{since } a \& b \text{ is 1 so}$$

$$\theta^{(Y+1)-1} (1-\theta)^{(N-Y+1)-1} \rightarrow \text{Let } A = Y+1$$

$$B = N-Y+1$$

$$\theta^{A-1} (1-\theta)^{B-1} \rightarrow \text{Beta PDF}$$

$$\theta|Y \sim \text{Beta}(Y+1, N-Y+1)$$

Problem 3

Posterior Plotting

```
theta <- seq(0, 1, by = 0.001)

data <- data.frame(
  Player = c("Stephen Curry", "Giannis Antetokounmpo", "John Wall", "LeBron James", "Kevin Durant"),
  overallProb = c(0.898, 0.770, 0.801, 0.674, 0.875),
  Y = c(24, 28, 66, 27, 13),
  N = c(26, 41, 82, 39, 16)
)

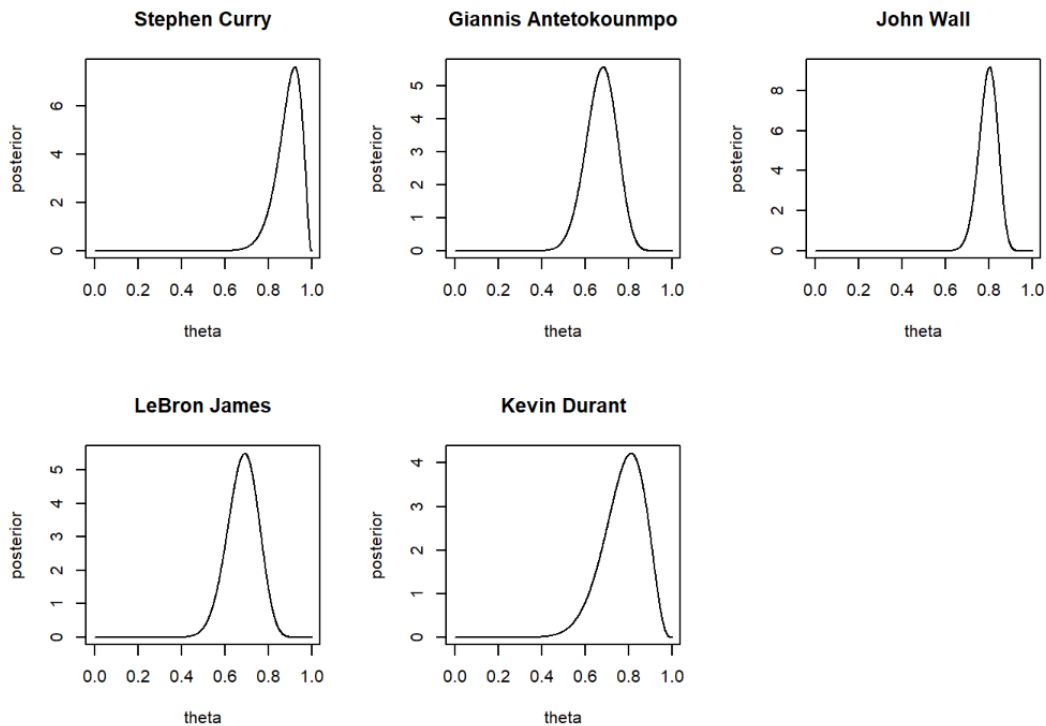
par(mfrow = c(2, 3))

for (x in 1:5) {
  posterior <- dbeta(theta, data[x, "Y"] + 1, data[x, "N"] - data[x, "Y"] + 1)
  plot(theta, posterior, type='l', main=data[x, "Player"])

  # Two-tailed test
  lower_tail <- pbeta(data[x, "overallProb"], data[x, "Y"] + 1, data[x, "N"] - data[x, "Y"] + 1)
  upper_tail <- 1 - pbeta(data[x, "overallProb"], data[x, "Y"] + 1, data[x, "N"] - data[x, "Y"] + 1)

  data$H_test_low <- lower_tail
  data$H_test_up <- upper_tail
}

data
```



```
## [1] "Stephen Curry"
## [1] 0.4706771
## [1] 0.5293229
## [1] ""
```

From the Hypothesis Testing above, given the prior and likelihood probability that the null hypothesis is true for $\theta < \text{Overall Performance}$ is 0.47 and for $\theta > \text{Overall performance}$ is 0.53.

```
## [1] "Giannis Antetokounmpo"  
## [1] 0.9166567  
## [1] 0.08334329  
## [1] ""
```

From the Hypothesis Testing above, given the prior and likelihood probability that the null hypothesis is true for $\theta < \text{Overall Performance}$ is 0.92 and for $\theta > \text{Overall performance}$ is 0.08.

```
## [1] "John Wall"  
## [1] 0.5092366  
## [1] 0.4907634  
## [1] ""
```

From the Hypothesis Testing above, given the prior and likelihood probability that the null hypothesis is true for $\theta < \text{Overall Performance}$ is 0.51 and for $\theta > \text{Overall performance}$ is 0.49.

```
## [1] "LeBron James"  
## [1] 0.4354453  
## [1] 0.5645547  
## [1] ""
```

From the Hypothesis Testing above, given the prior and likelihood probability that the null hypothesis is true for $\theta < \text{Overall Performance}$ is 0.44 and for $\theta > \text{Overall performance}$ is 0.57.

```
## [1] "Kevin Durant"  
## [1] 0.8457459  
## [1] 0.1542541  
## [1] ""
```

From the Hypothesis Testing above, given the prior and likelihood probability that the null hypothesis is true for $\theta < \text{Overall Performance}$ is 0.85 and for $\theta > \text{Overall performance}$ is 0.15.